By Nitesh Singh

Job Ready Free Live Training To Become A

SOC

Analyst L1 For Free

A SOC (Security Operations Center) is a dedicated facility operated by a specialized security team. This team aims to continuously monitor an organization’s network and resources and identify suspicious activity to prevent damage. This team works 24 hours a day, seven days a week.

Daily duties being a SOC Analyst

- Detect and prevent a data stealer infection on a coworker's laptop
- Analyze and stop a phishing campaign targeting the finance team
- Participate in a bigger incident, such as a full-scale ransomware attack
- Team up with your teammates to build detection rules and automations
- Go beyond cyber and understand how companies operate from the inside

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As a Tier-1 SOC Analyst, you are the first responder. You monitor, triage, escalate, and document every security alert.

- Monitor SIEM, EDR/XDR, firewall, email alerts
- Validate alerts: True Positive vs False Positive
- Escalate critical events to Tier-2/IR
- Log everything in ticketing systems

- L1 → L2 → L3 → IR → Blue Team / Threat Hunter

02 Core Concept

The main focus of the SOC team is to keep **Detection** and **Response** intact.

People, Process, and Technology coexist in a SOC environment. A team of professional individuals working on state-of-the-art security tools in the presence of proper processes is what makes a mature SOC environment.



Image Credit : tryhackme.com

03 People

- SOC Analyst (Level 1): Anything detected by the security solution would pass through these analysts first. These are the first responders to any detection. SOC Level 1 Analysts perform basic alert triage to determine if a specific detection is harmful. They also report these detections through proper channels.
- SOC Analyst (Level 2): While Level 1 does the first-level analysis, some detections may require deeper investigation. Level 2 Analysts help them dive deeper into the investigations and correlate the data from multiple data sources to perform a proper analysis.
- SOC Analyst (Level 3): Level 3 Analysts are experienced professionals who proactively look for any threat indicators and support in the incident response activities. The critical severity detection reported by Level 1 and Level 2 Analysts are often security incidents that need detailed responses, including containment, eradication, and recovery. This is where Level 3 analysts' experience comes in handy.
- Security Engineer: All analysts work on security solutions. These solutions need deployment and configuration. Security Engineers deploy and configure these security solutions to ensure their smooth operation.
- Detection Engineer: Security rules are the logic built behind security solutions to detect harmful activities. Level 2 and 3 Analysts often create these rules, while the SOC team can sometimes also utilize the detection engineer role independently for this responsibility.
- SOC Manager: The SOC Manager manages the processes the SOC team follows and provides support. The SOC Manager also remains in contact with the organization's CISO (Chief Information Security Officer) to provide him with updates on the SOC team's current security posture and efforts.

Source : tryhackme.com

Alert Triage

The alert triage is the basis of the SOC team. The first response to any alert is to perform the triage

Alert: Malware detected on Host: GEORGE PC

5 Ws Answers

What?

A malicious file was detected on one of the hosts inside the organization's network.

When?

The file was detected at 13:20 on June 5, 2024.

Where?

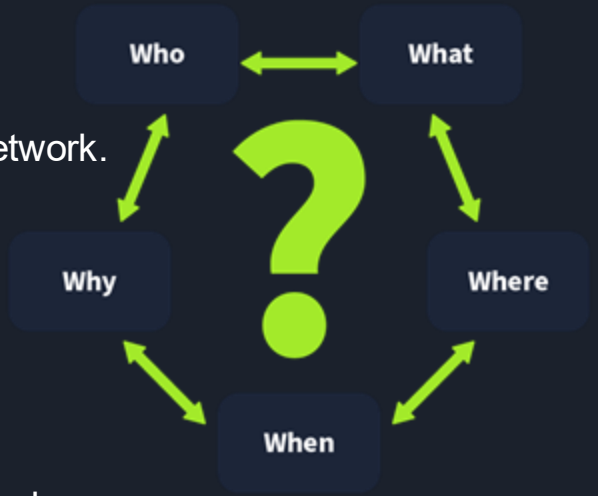
The file was detected in the directory of the host: "GEORGE PC".

Who?

The file was detected for the user George.

Why?

After the investigation, it was found that the file was downloaded from a pirated software-selling website. The investigation with the user revealed that they downloaded the file as they wanted to use a software for free.



Source : tryhackme.com

05 Process

Reporting

The detected harmful alerts need to be escalated to higher-level analysts for a timely response and resolution. These alerts are escalated as tickets and assigned to the relevant people. The report should discuss all the 5 Ws along with a thorough analysis, and screenshots should be used as evidence of the activity.

Incident Response and Forensics

Sometimes, the reported detections point to highly malicious activities that are critical. In these scenarios, high-level teams initiate an incident response.

A few times, a detailed forensics activity also needs to be performed. This forensic activity aims to determine the incident's root cause.

Technology

Having the right People and Processes in place would never be enough without security solutions for detection and response. The Technology portion in the SOC pillars refers to the security solutions. These security solutions efficiently minimize the SOC team's manual effort to detect and respond to threats.

Let's get a brief understanding of some of these security solutions:

SIEM: Security Information and Event Management (SIEM) is a popular tool used in almost every SOC environment. This tool collects logs from various network devices, referred to as log sources.

Detection rules are configured in the SIEM solution, which contains logic to identify suspicious activity. The SIEM solution provides us with the detections after correlating them with multiple log sources and alerts us in case of a match with any of the rules.

Modern SIEM solutions surpass this rule based detection analysis, providing us with user behavior analytics and threat intelligence capability. Machine learning algorithms support this to enhance the detection capabilities.

Source : tryhackme.com

Technology

Note: The SIEM solution only provides the Detection capabilities in a SOC environment.

EDR: Endpoint Detection and Response (EDR) provides the SOC team with detailed real-time and historical visibility of the devices' activities. It operates on the endpoint level and can carry out automated responses. EDR has extensive detection capabilities for endpoints, allowing you to investigate them in detail and respond with a few clicks.

Firewall: A firewall functions purely for network security and acts as a barrier between your internal and external networks (such as the Internet). It monitors incoming and outgoing network traffic and filters any unauthorized traffic. The firewall also has some detection rules deployed, which help us identify and block suspicious traffic before it reaches the internal network.

Several other security solutions play unique roles in a SOC environment, such as Antivirus, EPP (Endpoint Protection Platform), IDS/IPS, XDR (Extended Detection and Response), SOAR (Security Orchestration, Automation, and Response), and more. The decision on what Technology to deploy in the SOC comes after careful consideration of the threat surface and the available resources in the organization.

Source : tryhackme.com

08 Security Departments

In tiny companies, the IT department takes the role of securing the company. Small to medium-sized companies may have a generic "Information Security" team that does all sorts of tasks. Bigger companies have a CISO overseeing multiple security teams, each handling a specific task.

- **Red Team**: Offensive security experts, pentesters, or ethical hackers who look for security issues
- **GRC Team**: Specialists managing policies and ensuring compliance with regulations like PCI DSS
- **Blue Team**: Defensive security experts like SOC analysts, engineers, or incident responders

- **L1 Analysts**: Junior members who triage alerts and pass complex cases to L2
- **L2 Analysts:** Experienced members who investigate more advanced attacks
- **Engineers**: Experts in configuring security tools like EDR or SIEM
- **Manager**: A person who manages the whole SOC team

If SOC expertise is not enough or the incident goes out of control, you urgently call the "firefighters" - CIRT (Cyber Incident Response Team), also called CSIRT or CERT. The members should have a broad knowledge of cyber threats and handle breaches without depending on tools like EDR or SIEM.

Not every organization has the expertise to operate a SOC on its own and relies on a Managed Security Services Provider (MSSP), a company that delivers outsourced security services, most commonly SOC, to its clients. Working at MSSP is typically high-pressure, but it is also a good option to quickstart your career.

Why Humans Are Targeted

Humans are targeted because of the access they can provide - to websites, mailboxes, or databases. Some threat actors hunt for a specific access, while others are not so selective and just breach as many accounts as they can and decide how to use them later

Attacks targeting humans share a common trait: they rely on manipulating the victim into helping the attacker, whether knowingly or not. This tactic is known as social engineering, and it works by exploiting human psychology rather than technical flaws. For the tactic to succeed, it is designed to be:

Trustworthy: The attacker must appear legitimate so the victim trusts them
Emotional: The attack must trigger feelings like urgency, fear, or curiosity

How Humans Are Targeted

- Phishing Attacks
- Malware Downloads
- Deep Fakes
- Impersonation
- Other Attacks like Keylogging, Physical attacks etc

How To Defend Humans

Defending against threats involves two key tasks: Mitigation and Detection. Mitigation aims to prevent or reduce the chance and impact of attacks, for example by training employees or deploying a good anti-phishing solution.

How To Defend Humans

As a SOC analyst, your task is to detect and investigate attacks, because after sometime you will be tired of manual mitigation & will want automated way and there comes the SOC suite of automated solutions. Some solutions are :

- Anti-phishing solution
- Antivirus / EDR solution
- "Trust but verify" principle
- Security awareness training